

Project Assignment 1: Exploratory Survival Analysis

Spring 2026

DDS, Technion

Due: June 17, 2026 at 23:59 (submit in pairs via Moodle)

The goal of this first project assignment is to perform an initial exploratory survival analysis using a synthetic dataset derived from a study of patients hospitalized with acute decompensated heart failure. Between 2007 and 2017, 4590 patients were hospitalized at Rambam Health Care Campus with a primary diagnosis of heart failure and survived to discharge. The dataset contains follow-up information from hospital discharge until death or censoring, together with demographic, clinical, and laboratory characteristics (covariates). A description of the variables is available on Moodle.

Each pair of students will be randomly assigned a specific synthetic dataset generated using an individual random seed. Although the datasets differ across students, they are all based on the same underlying study and contain similar variables and survival outcomes. To protect patient privacy, a small number of observations were excluded during the anonymization and synthetic-data generation process. Consequently, the sample size in your assigned dataset may differ slightly from that of the original study.

The dataset assigned for this project, together with the research questions you formulate, will serve as the foundation for subsequent project assignments and for the final course project. Throughout the semester, you will apply increasingly sophisticated survival analysis methods to answer these research questions and compare the resulting conclusions.

Instructions

1. Load your assigned synthetic dataset and familiarize yourself with its structure. All analyses in this project should be performed using your assigned dataset.
2. Describe all covariates in the dataset:
 - Variable name, interpretation, and type (continuous, binary, categorical, ordinal, etc.)
 - Possible values or categories, and range for continuous covariates
3. Check for missing values in the covariates.
 - If missing values exist, describe the extent of missingness (e.g., percentage missing for each variable).
 - Handle missing data using simple imputation methods: replace missing values in continuous covariates with the overall median value, and replace missing values in categorical covariates with the mode (most frequent category).
4. Clearly define the event indicator and explain which observations are considered censored.

5. Report:
 - The sample size
 - The number and percentage of censored observations
6. Define the survival outcome:
 - What event constitutes a failure?
 - What is the failure time?
 - What is the time origin?
 - What are the time units (days, months, years, etc.)?
7. Describe the observed follow-up times:
 - Provide a summary of the observed time values (minimum, first quartile, median, third quartile, maximum).
 - Plot a histogram of the observed times.
 - Briefly comment on the distribution of the observed times.
8. Plot the Kaplan–Meier estimate of the overall survival function, together with confidence bands, and briefly interpret the curve.
9. Plot Kaplan–Meier curves (together with confidence bands) stratified by two or more meaningful categorical variables (for example, gender or history of comorbidities). Discuss differences in survival prognosis across groups.
10. Define one or more research questions motivated by the dataset. Examples:
 - Which patient characteristics are most strongly associated with survival outcomes?
 - Which method provides the most accurate prediction of survival probabilities?
 - Do different patient groups exhibit different survival patterns?
11. As a preliminary multivariable analysis, fit a Cox proportional hazards model using all available covariates. Report estimated hazard ratios, 95% confidence intervals, and p-values for all covariates. Briefly comment on which variables appear most strongly associated with the outcome. At this stage, the goal is exploratory rather than definitive inference; a more detailed regression analysis and comparison of survival models will be conducted in a subsequent project assignment.

Submission Guidelines

- Submit a short report (PDF) summarizing your analysis.
- Include all plots and relevant tables.

- Submit the code used for the analysis (Python).
- Clearly state any assumptions or preprocessing decisions.